HOMEWORK PROBLEM -1

QUES 1: Sum of two matrices

```
#include <stdio.h>
void main()
{
   int r1,c1,r2,c2;
   printf("enter no. of row and column of the matrix"); // m=no. of rows,n=no. of column.
   scanf("%d%d", &r1, &c1);
   int A[r1][c1];
   for (int i = 0; i < r1; i++)
   {
      for (int j = 0; j < c1; j++)
      {
         printf("enter element for 1st matrics");
         scanf("%d", &A[i][j]);
      }
   }
   for (int i = 0; i < r1; i++)
   {
      for (int j = 0; j < c1; j++)
      {
         printf("%d", A[i][j]);
      }
      printf("\n");
   }

   int B[r2][c2];
   for (int i = 0; i < r2; i++)
   {
      for (int j = 0; j < c2; j++)
      {
```

```
        printf("enter elements for 2nd matrics");
        scanf("%d", &B[i][j]);
    }

}
for (int i = 0; i < r2; i++)
{
    for (int j = 0; j < c2; j++)
    {
        printf("%d", B[i][j]);
    }
    printf("\n");
}
printf("The addition of the two matrices is:\n");
int C[r1][c1];
for (int i = 0; i < r1; i++)
{
    for (int j = 0; j < c1; j++)
    {
        C[i][j] = A[i][j] + B[i][j];
    }

}
for (int i = 0; i < r1; i++)
{
    for (int j = 0; j < c1; j++)
    {
        printf("%d",C[i][j]);
    }
    printf("\n");
}
```

QUES 2: Write a program in c to read n numbers of values in an array display them in reverse order

```
#include<stdio.h>
int main()
{
   int n;
   printf("enter the number of elements: ");
   scanf("%d",&n);
   int arr[n];
   printf("enter elements");
   for(int i=0;i<n;i++)
   {
      scanf("%d",&arr[i]);
   }
   printf("the reversed array is: ");
   for(int i=n-1;i>=0;i--)
   {
      printf("%d ",arr[i]);
   }
   return 0;
}
```

QUES 3: Count the total number of duplicate elements in an array

```
#include <stdio.h>
void swap(int *a, int *b)
{
   int temp = 0;
   temp = *a;
   *a = *b;
   *b = temp;
}
```

```
void bubble(int arr[], int n)
{
   int i, j;
   for (i = 0; i < n - 1; i++)
   {

      for (j = 0; j < n - 1; j++)
      {

         if (arr[j] > arr[j + 1])
         {
            swap(&arr[j], &arr[j + 1]);
         }
      }
   }
}
int count(int arr[],int n)
{
   int count=0;
   for(int i=0;i<n-1;i++)
   {
      if(arr[i]==arr[i+1])
      count++;
   }
   return count;
}
int main()
{
   int arr[] = {7, 6, 7, 5, 3, 2, 5, 6, 5};
   int size = sizeof(arr) / sizeof(arr[0]);
   bubble(arr, size);
```

```
    for (int i = 0; i < size; i++)
    {
        printf("%d ", arr[i]);
    }
    int result=count(arr,size);
    printf("\nthe number of duplicate elements is: %d",result);
    return 0;
}
```

QUES 4: Print unique element in an array

```
#include <stdio.h>
int main()
{
    int arr[] = {7, 6, 7, 5, 3, 2, 5, 6, 5, 0, 9};
    int size = sizeof(arr) / sizeof(arr[0]);
    int i, j, count = 0;
    for (i = 0; i < size; i++)
    {
        count = 1;
        for (j = 0; j < size; j++)
        {
            if (arr[i] == arr[j] && i != j)
                count = 0;
        }
        if (count == 1)
            printf("%d ", arr[i]);
    }

    return 0;
}
```

QUES 5: Insert in sorted array

```
#include <stdio.h>
#include<stdbool.h>
void insert(int arr[], int n, int ele)
{
   bool find =0;
   int pos=0;
   for (int i = 0; i < n - 1; i++)
   {
      if (ele > arr[i] && ele <= arr[i + 1])
      {
         pos = i + 1;
         find=1;
         break;
      }

   }
   if(find==1)
   {
      for(int j=n;j>pos;j--)
      {
         arr[j]=arr[j-1];
      }
      arr[pos]=ele;
   }
   else
   arr[n]=ele;
}

int main()
{
```

```
    int arr[] = {2, 3, 4, 5, 6, 7, 7, 8, 9};
    int size=sizeof(arr)/sizeof(arr[0]);
    int key;
    printf("enter the element to insert: ");
    scanf("%d", &key);

    insert(arr, size, key);
    for (int i = 0; i <= size; i++)
    {
        printf("%d ", arr[i]);
    }
    return 0;
}
```

QUES 6: Insert in unsorted array

```
#include <stdio.h>
void insert(int arr[], int n, int ele,int pos)
{
    for(int i=n;i>pos;i--)
    {
        arr[i]=arr[i-1];
    }
    arr[pos]=ele;
}

int main()
{
    int size;
    printf("enter thr size of the array: ");
    scanf("%d",&size);
```

```
    int arr[size];
    printf("enter the elements: ");
    for(int i=0;i<size;i++)
    {
        scanf("%d",&arr[i]);
    }

    int key,pos;
    printf("enter the element to insert: ");
    scanf("%d", &key);
    printf("enter the position for the element to be inserted: ");
    scanf("%d", &pos);
    insert(arr, size+1, key,pos);
    for (int i = 0; i <= size; i++)
    {
        printf("%d ", arr[i]);
    }
    return 0;
}
```

QUES 7: Matrix multiplication

```
#include <stdio.h>
void main()
{
    int r1, c1, r2, c2;
    printf("enter no. of row and column of the matrix 1: \n"); // m=no. of rows,n=no. of column.
    scanf("%d%d", &r1, &c1);
    int A[r1][c1];
    printf("enter elements for 1st matrix\n");
    for (int i = 0; i < r1; i++)
```

```
{

   for (int j = 0; j < c1; j++)
   {

      scanf("%d", &A[i][j]);
   }
}
printf("the first matrix is: \n");
for (int i = 0; i < r1; i++)
{
   for (int j = 0; j < c1; j++)
   {
      printf("%d", A[i][j]);
   }
   printf("\n");
}
printf("enter no. of row and column of the matrix2: \n"); // m=no. of rows,n=no. of column.
scanf("%d%d", &r2, &c2);
int B[r2][c2];
printf("enter elements for 2nd matrix: \n");
for (int i = 0; i < r2; i++)
{

   for (int j = 0; j < c2; j++)
   {

      scanf("%d", &B[i][j]);
   }
}
printf("the second matrix is: \n");
```

```
    for (int i = 0; i < r2; i++)
    {
        for (int j = 0; j < c2; j++)
        {
            printf("%d", B[i][j]);
        }
        printf("\n");
    }
    printf("The multiplication of the two matrices is:\n");
    int E[r1][c2];
    for (int i = 0; i < r1; i++)
    {
        for (int j = 0; j < c2; j++)
        {
            E[i][j]=0;
            for(int k=0;k<c1;k++)
            {
                E[i][j]=E[i][j]+(A[i][k]*B[k][j]);
            }
        }

    }
    for (int i = 0; i < r1; i++)
    {
        for (int j = 0; j < c2; j++)
        {
            printf("%d ",E[i][j]);
        }
        printf("\n");
    }
}
```

QUES 8: Matrix transpose

```
#include <stdio.h>
void main()
{
   int n;
   printf("enter no. of rows and column it should be a square matrix");
   scanf("%d", &n);
   int arr[n][n];
   int temp;
   for (int i = 0; i < n; i++)
   {
      printf("enter elements of row %d", i + 1);
      for (int j = 0; j < n; j++)
      {
         scanf("%d", &arr[i][j]);
      }
      printf("\n");
   }
   printf("your input was\n");
   for (int i = 0; i < n; i++)
   {

      for (int j = 0; j < n; j++)
      {
         printf("%d ", arr[i][j]);
      }
      printf("\n");
   }
   for (int i = 0; i < n; i++)
   {
```

```
        for (int j = i; j < n; j++)//or,j=0;j<=i,j++
        {
            temp = arr[i][j];
            arr[i][j] = arr[j][i];
            arr[j][i]=temp;
        }
    }
    printf("the tranpose of the matrix is \n");
    for (int i = 0; i < n; i++)
    {

        for (int j = 0; j < n; j++)
        {
            printf("%d ", arr[i][j]);
        }
        printf("\n");
    }
}
```

QUES 9: Right diagonal sum

```
#include<stdio.h>
int main()
{
    int r, c;
    printf("enter no. of rows and columns of the matrix: \n"); // m=no. of rows,n=no. of column.
    scanf("%d%d", &r, &c);
    int A[r][c];
    printf("enter elements for the matrix\n");
    for (int i = 0; i < r; i++)
    {
```

```
        for (int j = 0; j < c; j++)
        {

            scanf("%d", &A[i][j]);
        }
    }
    printf("the matrix is: \n");
    for (int i = 0; i < r; i++)
    {
        for (int j = 0; j < c; j++)
        {
            printf("%d ", A[i][j]);
        }
        printf("\n");
    }
    printf("\nthe sum of the right diagonal elements is: ");
    int sum=0;
    for(int i=0;i<r;i++)
    {
        for(int j=c-i-1;j>=0;j--)
        {

            sum+=A[i][j];
            break;
        }
    }
    printf("%d",sum);
    return 0;
}
```

QUES 10: Check for identity matrix

```
#include<stdio.h>
#include<stdbool.h>
int main()
{
   int r1, c1;
   printf("enter no. of row and column of the matrix: \n"); // m=no. of rows,n=no. of column.
   scanf("%d%d", &r1, &c1);
   int A[r1][c1];
   bool find=true;
   printf("enter elements for matrix\n");
   for (int i = 0; i < r1; i++)
   {

      for (int j = 0; j < c1; j++)
      {

         scanf("%d", &A[i][j]);
      }
   }

   for (int i = 0; i < r1; i++)
   {
      for (int j = 0; j < c1; j++)
      {
         if(A[i][i]==1)
         {
            find=true;
         }
         else
         find=false;
```

```
        }
    }
    if(!find)
    {
        printf("matrix is not a identity matrix");
    }
    else
    printf("identity matrix");
    return 0;
}
```

QUES 11: To find number of vowels,consonants,digits and white spaces

```
#include <stdio.h>

int main()
{
    char str[100];
    int vowels = 0, consonants = 0, digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for (int i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
            str[i] == 'o' || str[i] == 'u' ||
            str[i] == 'A' || str[i] == 'E' || str[i] == 'I' ||
            str[i] == 'O' || str[i] == 'U')
        {
            vowels++;
```

```
        }
        else if ((str[i] >= 'a' && str[i] <= 'z') ||
                 (str[i] >= 'A' && str[i] <= 'Z'))
        {
            consonants++;
        }
        else if (str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if (str[i] == ' ' || str[i] == '\t')
        {
            spaces++;
        }
    }

    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);
    printf("Digits = %d\n", digits);
    printf("White spaces = %d\n", spaces);

    return 0;
}
```

QUES 12: Frequency of a character in a string

```
#include <stdio.h>

int main()
{
    char str[100];
```

```
    char ch;
    int count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter the character: ");
    scanf("%c", &ch);

    for (int i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ch)
        {
            count++;
        }
    }

    printf("Frequency of '%c' = %d\n", ch, count);

    return 0;
}
```

QUES 13: Count words in a string

```
#include <stdio.h>

int main()
{
    char str[100];
    int count = 0;
```

```
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for (int i = 0; str[i] != '\0'; i++)
    {
        if (str[i] != ' ' && (str[i + 1] == ' ' || str[i + 1] == '\0'))
        {
            count++;
        }
    }

    printf("Number of words = %d\n", count);

    return 0;
}
```

QUES 14: Sort string array

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[5][20] = {
        "banana",
        "apple",
        "orange",
        "mango",
        "grapes"
    };
```

```
    char temp[20];

    for (int i = 0; i < 5 - 1; i++)
    {
        for (int j = i + 1; j < 5; j++)
        {
            if (strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("Sorted strings:\n");

    for (int i = 0; i < 5; i++)
    {
        printf("%s\n", str[i]);
    }

    return 0;
}
```

QUES 15: Toggle case of a sentence (lower case to upper case and vice versa)

```
#include <stdio.h>

int main()
```

```
{
    char str[100];

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    for (int i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
        else if (str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
    }

    printf("Toggled sentence: %s", str);

    return 0;
}
```